

Session 4 · Tuesday 8 September 2026 · Two in the room, two on the problem set. Start wherever the scaffolding stops helping you.

Where a regulation function comes from

Four versions of one derivation, with **less of my working shown each time**. We do items 1 and 2 in the room — ten minutes, and item 1 is already done. Items 3 and 4 are the same technique with the scaffolding gone; they are on **PS1**, and they are there rather than here because they take longer than ten minutes and this room is worth more than that.

The **bold labels** say what each step *accomplishes*. At each ► **Why does that step follow?**, answer in writing before going on.

NOTATION	$x = [X]$, the free transcription factor	SITE	P = empty promoter, PX = bound
DISSOCIATION CONSTANT	$K_d = [P][X]/[PX]$ — large K_d means weak binding	ASSUME	binding equilibrates far faster than transcription

1 • Fully worked — one site

A transcription factor X binds a single site on a promoter. What fraction of promoters are bound, as a function of $[X]$?

Write the equilibrium, and say what is held fixed.



X is buffered: there is far more of it than there is promoter, so binding does not deplete it. That is why x is a parameter and not a variable.

Form the fraction — the thing you actually want, not the concentration.

$$f_{\text{bound}} = \frac{[PX]}{[P] + [PX]} = \frac{[P] x / K_d}{[P] + [P] x / K_d}$$

Cancel the thing you cannot measure.

$$f_{\text{bound}} = \frac{x / K_d}{1 + x / K_d} = \frac{x}{K_d + x}$$

Worth carrying: that is a Hill function, with $n = 1$ and $K = K_d$. The half-point is at $x = K_d$ — which is what a dissociation constant means.

► **Why does that step follow?** $[P]$ cancelled. Why is that not an accident of this problem — what would have to be true for it *not* to cancel?

2 • The last step is yours — n sites, all or nothing

Now the promoter has n identical sites, and cooperativity is so strong that only two states are ever populated: **empty**, or **fully occupied**. Nothing in between.

Write the equilibrium for the idealised two-state promoter.



Now you: form the fraction, cancel, and put it in Hill form — that is, identify the K for which your answer reads $x^n / (K^n + x^n)$.

► **Why does that step follow?** Your K is not K_d . Say in one sentence what K is, and what it would mean physically for two promoters with the same K to have different K_d .

► **And the honest question.** "No partially bound states" is a lie. Whose lie is it — what physical situation makes it a *good* lie, and what would you measure to find out whether it is one here?

On the problem set — the same technique, unscaffolded

These are **PS1 Q4b and Q5**. They are not homework in the sense of "more of the same"; each one is the place where the idealisation you just used gets tested.

3 • The last two steps are yours — what n actually buys you

How much must x change to take the promoter from 10% to 90% occupied?

State what you are solving. Set $x^n/(K^n + x^n)$ equal to 0.1, and again to 0.9, and solve each for x . Do it **algebraically** — no numerical search.

Now you: everything else. You should find the ratio x_{90}/x_{10} depends only on n , not on K . Report it as a formula, then evaluate it for $n = 1, 2, 4, 8$.

This number is the entire practical content of a Hill coefficient. A circuit that must switch cleanly needs a small fold-change here; that requirement is what sends people looking for cooperativity in the first place.

4 • Bare problem — BioE 247; 147 may attempt it for no credit

*Relax the idealisation. Two **independent, identical** sites — binding at one does nothing to the other — each with dissociation constant K_d .*

Write the partition function over all four states (empty, site 1 only, site 2 only, both), derive the fractional occupancy, and put it in Hill form.

State your assumptions as you go. Two things to be careful about, both of which are the actual question rather than a technicality:

- **Say which fraction you are computing.** "Fraction of sites occupied" and "fraction of promoters with at least one site bound" are different functions of x , and only one of them is a Hill function at all.
- **Then state what your answer implies** for reading a measured Hill coefficient off a dose-response curve, and what would have to be true of the binding energetics to recover $n = 2$.

A Hill coefficient is not a count of binding sites. It is a measure of cooperativity, and the number of sites is only its ceiling — a bound you reach in a limit that nothing in a cell quite occupies.